```
from google.colab import files
uploaded = files.upload()
```

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```
Saving CO2 Emissions_Canada.csv to CO2 Emissions_Canada (1).csv
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
```

```
# Load dataset
df = pd.read_csv("CO2 Emissions_Canada.csv")

# Display first 5 rows
df.head()
```

| | Make | Model | Vehicle Class | Engine Size(L) | Cylinders | Transmission | Fuel Type | Fuel Consumption City (L/100 km) | Fuel Consumption Hwy (L/100 km) | Fuel Consumption Comb (L/100 km) | Fuel Consumption Comb (mpg) | CO2 Emissions(g/km) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | ACURA | ILX | COMPACT | 2.0 | 4 | AS5 | Z | 9.9 | 6.7 | 8.5 | 33 | 196 |
| **1** | ACURA | ILX | COMPACT | 2.4 | 4 | M6 | Z | 11.2 | 7.7 | 9.6 | 29 | 221 |
| **2** | ACURA | ILX HYBRID | COMPACT | 1.5 | 4 | AV7 | Z | 6.0 | 5.8 | 5.9 | 48 | 136 |
| **3** | ACURA | MDX 4WD | SUV - SMALL | 3.5 | 6 | AS6 | Z | 12.7 | 9.1 | 11.1 | 25 | 255 |
| **4** | ACURA | RDX | SUV - | 3.5 | 6 | AS6 | Z | 12.1 | 8.7 | 10.6 | 27 | 244 |

```
# Feature and Target
X = df[['Engine Size(L)']]   # Feature
y = df['CO2 Emissions(g/km)']  # Target
```

```
# Fit Linear Regression
model = LinearRegression()
model.fit(X, y)

# Extract coefficient (β₁)
beta_1 = model.coef_[0]

print("Original Regression Coefficient (β₁):", beta_1)
```

```
Original Regression Coefficient (β₁): 36.77731518641943
```

```
B = 2000
n = len(df)

bootstrap_betas = []
```

```
for i in range(B):
    # Resample with replacement
    sample = df.sample(n, replace=True)

    X_sample = sample[['Engine Size(L)']]
    y_sample = sample['CO2 Emissions(g/km)']

    model = LinearRegression()
    model.fit(X_sample, y_sample)

    bootstrap_betas.append(model.coef_[0])

bootstrap_betas = np.array(bootstrap_betas)
```

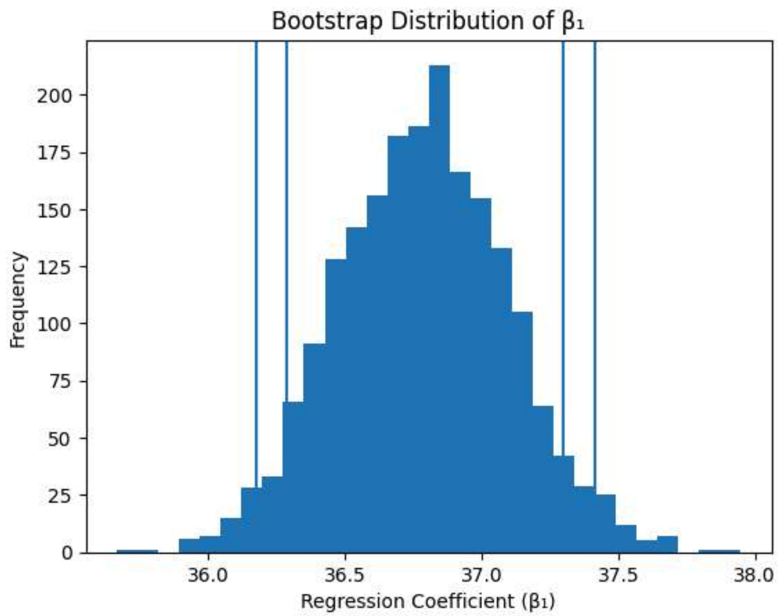
```
# 90% Confidence Interval
ci_90 = np.percentile(bootstrap_betas, [5, 95])

# 95% Confidence Interval
ci_95 = np.percentile(bootstrap_betas, [2.5, 97.5])

print("90% Confidence Interval for β₁:", ci_90)
print("95% Confidence Interval for β₁:", ci_95)
```

```
90% Confidence Interval for β₁: [36.28634704 37.29681204]
95% Confidence Interval for β₁: [36.17816778 37.41577767]
```

```
plt.hist(bootstrap_betas, bins=30)
plt.axvline(ci_90[0])
plt.axvline(ci_90[1])
plt.axvline(ci_95[0])
plt.axvline(ci_95[1])
plt.title("Bootstrap Distribution of β₁")
plt.xlabel("Regression Coefficient (β₁)")
plt.ylabel("Frequency")
plt.show()
```




The bootstrap 90% and 95% confidence intervals for the regression coefficient $\beta_1$ do not include zero. Therefore, engine size significantly affects $CO_2$ emissions. The 95% confidence interval is wider than the 90% interval, as expected